

Mohamed Mostafa Hassan Afify

Data Analyst | AI Engineer

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PROFESSIONAL SUMMARY

Data & AI professional working at the intersection of **analytics** and **applied AI**. Turns raw data into decisions with **SQL**, **Power BI**, **Tableau** and **Python** — recently tracing **\$7.8M** of at-risk revenue to its root causes — and builds production-minded AI systems including agentic **RAG** pipelines, multi-agent workflows and real-time computer vision models. Currently completing the **NTI Creativa Data Analysis & Freelancing Diploma** (Digital Egypt Youth Initiative). Trained by **NVIDIA & ITI**, **Google**, **Route Training Center** and **Instant & Orange Digital Center (ODC)**.

EDUCATION

B.Sc. in Computers and Management Information Systems (MIS)

El Gazeera Higher Institute, Cairo

2019 – 2023

PROFESSIONAL EXPERIENCE

Maintenance Engineer — Vision Company, Cairo

Dec 2023 – Jan 2025

- Tracked equipment performance and maintenance logs to surface recurring failure patterns, turning scattered records into a usable reporting base.
- Audited inventory and sales data for inconsistencies, resolving data-quality issues that were distorting operational reports.
- Structured technical and procurement data to support performance tracking and day-to-day operational decisions.

PROJECTS

IBM Telco Customer Churn — End-to-End Analytics

[GitHub](#)

- Modeled 5 relational tables (7,043 customers) into a **SQL Server** star schema; wrote 15 T-SQL queries (JOIN, CASE, HAVING, window functions) and built a **Python** cleaning pipeline with leakage removal.
- Designed 4 connected **Tableau** dashboards; found month-to-month contracts churn **18×** more than two-year (45.8% vs 2.5%), driving 5 recommendations on **\$7.8M** of at-risk lifetime value.

NTI Creativa Data Analysis & Freelancing Diploma

Wide World Importers — Power BI Sales Analytics

[GitHub](#)

- Rebuilt a broken data model (26,397 transactions) into a corrected star schema (1 fact + 5 dimensions); authored **24+ DAX measures** and designed 4 report pages (~63 visuals) via **Power Query**.

NTI Creativa Data Analysis & Freelancing Diploma

Ames Housing — Data Cleaning & Preprocessing

[GitHub](#)

- Handled missing values, eliminated data leakage, encoded categorical features and applied log transformations to correct skewness; validated with before/after EDA visualizations.

NTI Creativa Data Analysis & Freelancing Diploma

Agentic RAG Deep Visualizer

[GitHub](#)

- Stateful agentic RAG system (**LangGraph** + **Groq Llama 3.3 70B**) with a self-correcting dual-memory loop: checks a local **Qdrant** store first, then falls back to live web scraping (**Firecrawl**) and auto-ingests new knowledge.

Route Generative AI Diploma

Multi-Agent Deep Researcher

[GitHub](#)

- Designed a 4-agent workflow (**Planner**, **Researcher**, **Writer**, **Critic**) on **LangGraph** + **Groq Llama 3.1**; used **Redis** as a persistent state checkpointer for fault-tolerant sessions.

Route Generative AI Diploma

Multimodal Vision OCR Agent

[GitHub](#)

- Fully local, cloud-free Vision-Language OCR pipeline (**Qwen2.5-VL 3B** via **Ollama**, **FastAPI**) with an **OpenCV** stage (deskewing, CLAHE, patch-aligned resizing) producing structured JSON.

Route Generative AI Diploma

RAG Pipeline Evaluation & Quality Observability

[GitHub](#)

- Metrics-driven RAG evaluation engine using **Ragas** with **GPT-4o-mini** as LLM-as-a-Judge, auditing a live **Qdrant** + **Groq Llama 3.3** pipeline on Context Precision, Context Recall, Faithfulness and Answer Relevancy.

Route Generative AI Diploma

GroundedRAG — Safe & Evaluated Document Q&A System

[GitHub](#) | [Demo](#)

- Full RAG pipeline (ingestion, embedding, retrieval, generation) with local LLM inference and automated hallucination evaluation.

NVIDIA DLI Generative AI Training — Advanced Level

[GitHub](#) | [Demo](#)

Real-Time Fire & Smoke AI Guard

- YOLO detection model reaching **0.739 mAP@0.5** and **86.5% recall**; Streamlit interface for real-time hazard analysis.

Computer Vision Training — Instant & ODC

[GitHub](#) | [Demo](#)

COVID-19 X-ray Classification

- MobileNetV2 transfer learning reaching **96.5% accuracy** distinguishing COVID-19 from pneumonia on chest X-rays.

Route AI & Machine Learning Diploma

[GitHub](#) | [Demo](#)

Object Tracking System

- Multi-object tracking using OpenCV motion analysis and feature matching for real-time video streams.

Route AI & Machine Learning Diploma

[GitHub](#) | [Demo](#)

OCR Document Intelligence App

- Tesseract + OpenCV ID scanner; custom preprocessing improved extraction accuracy by **25%**.

Route AI & Machine Learning Diploma

CERTIFICATIONS & TRAINING

NTI Creativa — Data Analysis & Freelancing Diploma — Cairo University (Digital Egypt Youth Initiative)

Ongoing

Generative AI Diploma — Route Training Center | 4 Courses + Capstone Projects

[View Certificate](#)

Google AI Professional Certificate — Google / Coursera | 7 Courses

[Verify Credential](#)

NVIDIA DLI — Generative AI Training (Advanced Level) — ITI & NVIDIA | 2 Courses + Capstone

Computer Vision Training — Instant & Orange Digital Center (ODC) | 2 Projects

[View Certificate](#)

NVIDIA DLI — Generative AI Training (Beginner Level) — ITI & NVIDIA | 5 Courses

[View Certificate](#)

Database Fundamentals — Mahara Tech (ITI)

[View Certificate](#)

AI & Machine Learning Diploma — Route Training Center

[View Certificate](#)

TECHNICAL SKILLS

- Data Analysis & BI:** SQL Server (SSMS, T-SQL, Window Functions, Star Schema), Power BI (DAX, Power Query), Tableau, Excel, Pandas, NumPy, Data Cleaning & Preprocessing, Statistics, EDA, Data Visualization
- Generative AI & Agents:** LLMs, RAG, LangChain, LangGraph, CrewAI, Agentic Workflows, Prompt Engineering, Vector Search (Qdrant, ChromaDB, Pinecone), Redis, Groq, Ollama, OpenAI API, Gemini API, Hugging Face
- Computer Vision & VLMs:** YOLO, OpenCV, Qwen2.5-VL, Object Detection, Image Classification, OCR (Tesseract), Real-Time Video Analysis, Diffusion Models, CUDA
- Deep Learning & ML:** CNN, Transformers, Transfer Learning, TensorFlow, Scikit-learn, Feature Engineering, Model Evaluation
- Evaluation & APIs:** Ragas, LLM-as-a-Judge, Faithfulness / Context Precision & Recall, FastAPI, Streamlit, Pydantic, Asyncio, Git, Docker
- Soft Skills:** Problem Solving, Critical Thinking, Fast Learner, Attention to Detail, Communication, Team Collaboration, Adaptability

LANGUAGES

Arabic (Native) • English (Professional Working Proficiency)